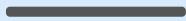




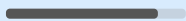
✉ constantin.riff@gmail.com
🏠 Lausanne, Switzerland
☎ +41 78 447 43 92
🇫🇷 French

Languages

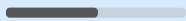
French



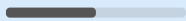
English



German



Italian



Activities and Passions

Travelling

Backpacking, Road trip through many countries to discover cultures and landscapes

Sports :

Hiking, Running, Ski Touring, Alpinism

Music :

Playing the piano for many years, listening to all types of music, participating in jam session

Social networks

in @Constantin Riff

Constantin Riff

Mechanical Engineering & AI Graduate from ETH Zurich

Education

Master Degree in Mechanical engineering

From September 2022 to September 2025 **ETHZ** Zürich

- **Relevant coursework:** Mathematical Optimization, Machine Learning, Computer Vision, Finite Element Method II, Imitation and Reinforcement Learning, Big Data Analytics, Vehicle Propulsion Systems, Carbon Capture Technologies, Autonomous Robot Planning, Robot Dynamics, Robotic Systems, Rehabilitation Engineering

Research project: @Laboratory of Biosensors and Bioelectronics (LBB): "Developing microfluidics for at-home blood testing".

- Competencies: Prototyping, 3D Printing, CAD (Fusion & Adobe Illustrator), R&D within a team, electrochemical assay.

Master's Thesis: @CREATE Lab (EPFL) - Embodied AI for World-Aware Robotics

- Designed a perception-reasoning-action framework where a large language model predicts physical outcomes and plans robot actions from visual observations.
- Built a closed-loop system controlling a UR3 robotic arm using LLM-generated action primitives.
- Implemented short-term and long-term memory mechanisms to maintain temporal consistency during multi-step tasks.
- Evaluated physical reasoning capabilities across tasks involving mechanics, object stability, and fluid dynamics.
- Identified failure modes arising from perception ambiguity and visual grounding.

Bachelor's degree in Mechanical Engineering

From 2019 to 2022 **École Polytechnique Fédérale de Lausanne (EPFL)**, Lausanne

2021-2022 : Exchange year at *La Sapienza*, in Rome

Subjects studied: Mathematical analysis, Numerical Analysis, Physics, Fluid Mechanics, Thermodynamics, Electromagnetism, Material Science, Finite Element Method

Scientific Baccalauréat

2019 , **Institution des Chartreux**, Lyon

with *Très Bien* mention and final grade 17.10/20 Scientific program with specialization in Physics.

Employment and Internships

Teaching Mentor : weekly support sessions for first-year students in math and physics.

Since February 2021 **EPFL** Lausanne

Engineer Internship at Flexis in R&D

From September 2024 to February 2025 **Flexis** Paris

Flexis, a Renault-Volvo joint venture developing a new generation of electric utility vehicles for last-mile delivery. I worked daily within the 3DEXPERIENCE PLM environment (CATIA, ENOVIA), managing product data and collaborating with design, machining and materials engineering teams on real product decisions. I designed a data-driven framework to assess and reduce the CO₂ footprint of next-generation vehicles, providing quantitative input to engineering and component sourcing decisions. I also led the internal adoption of Microsoft Copilot across the company, training colleagues from non-technical backgrounds to integrate a new AI tool into their daily workflows. In parallel, I contributed to the ESG workstream, supporting C-level on sustainability requirements, translated technical results for the commercial and communication teams, and supported a successful BPI France grant application by communicating engineering value to a non-technical investor board.

Internship at the European Organization for Nuclear Research

From May 2018 to June 2018 **CERN** Genève

Discovery of the daily job of a Physicist with Dr. David Francis and Dr. Benedetto Gorini and also the daily life of an Engineer with Didier Lombard. Discovery of the fields of particle physics, through fundamental research and the applications for the society.

Skills and Technologies

- **Programming languages:** C, C++, Python (scikit-learn, PyTorch, TensorFlow, pandas, NumPy, matplotlib), LaTeX, SQL
- **Frameworks and Softwares:** ROS 2, Git, Matlab, Docker, LabVIEW, Microsoft Office, CATIA, Fusion 360, SAP 2000, IsaacLab

Awards and Competitions

- **First place at the Act for Change Lab [2nd semester of the first year at EPFL, 2020] :** Competition with study of a case linked to energy. With my group of four other EPFL-students, we studied *Urbanization of data centers and energy recovery*.
- **General mathematics Contest [2019]:** Rank: 1st in my high school.